

HCSM-38

Hidden Sector Content of the Hodge Complex Standard Model

The $\lambda = 4$ Hodge Lift, the D_4 Decomposition of the Complement, and the Derivation of the Hidden $5 + 6 + 31$ Content

Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model (HCSM) identifies the 14 scalar modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04, HCSM-23). This paper establishes the hidden sector content of the framework: the 42 modes at $\lambda = 4$ that appear in the *Hodge lift* $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$ but are not identified with the 14 visible scalar modes. We compute the full D_4 decomposition of the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ from character theory, derive the D_4 content of the hidden sector as

$$6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E,$$

and show that under the sector-coarse correspondence of HCSM-06 this content maps to the physical hidden sector

$$(d+1) \text{ weak doublets} + (d+2) \text{ color triplets} + \left(\frac{N}{2} - 1\right) \text{ singlets} = 5 + 6 + 31.$$

The hidden sector content is derived from the D_4 representation theory of the Hodge complex on Λ . Three physical choices are disclosed: the identification of D_4 irreps with $SU(3) \times SU(2) \times U(1)$ representations, the $U(1)_Y$ neutrality of the hidden singlets, and the fermionic statistics of the hidden B_1 doublets. These three choices are consistent with the HCSM-06/07 disclosure pattern. The result sets up the Weinberg angle derivation of companion paper HCSM-39. Every numerical claim is verified in reproducible Python.

Derivation status summary.

- **DERIVED:** the full D_4 decomposition of the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ from character theory; the D_4 content of the Hodge lift at $\lambda = 4$; the identification of the visible sector with $\Omega^0|_{\lambda=4}$; the hidden sector as the complement $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$; the D_4 content $6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E$; the physical content $(d+1) + (d+2) + (N/2 - 1) = 5 + 6 + 31$.
- **DISCLOSED INPUTS:** (i) the identification of D_4 irreps with $SU(3) \times SU(2) \times U(1)$ representations via the sector-coarse correspondence (HCSM-06); (ii) $U(1)_Y$ neutrality of the hidden singlets; (iii) fermionic statistics of the hidden B_1 doublets.
- **OPEN:** the microscopic derivation of the three physical choices from the substrate; the exact $U(1)_Y$ charges of the hidden color triplets and doublets; the contribution of the hidden sector to the running of the gauge couplings below the HCSM scale.

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1 Introduction

1.1 The visible sector and its completion

The Hodge Complex Standard Model identifies the 14 scalar modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with one Standard Model generation (HCSM-04, HCSM-23). These 14 modes are the *visible sector*: they are identified with the SM particles

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t),$$

and every downstream HCSM result the charge theorem, the mass map, the CKM and PMNS matrices, the graviton is a statement about them.

The Hodge complex on Λ has total dimension $64 + 128 + 64 = 256$. Of these, only 56 lie in the $\lambda = 4$ eigenspace of the Hodge–Dirac operator $D = d + d^*$ (HCSM-02). The remaining 200 modes are the bath of the self cancellation theorem (HCSM-10).

The 56-dimensional $\lambda = 4$ sector itself splits by form degree:

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 14 + 28 + 14.$$

The 14 scalar modes in Ω^0 are the visible sector. The remaining 42 modes at $\lambda = 4 - 28$ in Ω^1 and 14 in Ω^2 are the *hidden sector* in the sense of this paper.

This is the framework’s structural statement: the visible SM content sits in the scalar sector, and the Hodge lift of the same multiplet into the vector and pseudo-scalar sectors provides additional, structurally necessary content.

1.2 The one line summary

The visible sector is the scalar shadow of the $\lambda = 4$ multiplet. The hidden sector is the Hodge lift.

1.3 The hidden sector question

The visible sector is a 14-mode multiplet. If the Hodge complex is a faithful substrate, then the Hodge lift of these 14 modes into the vector and pseudo-scalar sectors must be interpreted. This paper answers the question: what is the content of the hidden sector, and how does it transform under D_4 ?

The answer, derived in Sections 4–6, is

$$\text{hidden content} = 5 \text{ weak doublets} + 6 \text{ color triplets} + 31 \text{ singlets},$$

or, in terms of the framework constants $d = 4$ and $N = 64$,

$$\text{hidden content} = (d + 1) + (d + 2) + \left(\frac{N}{2} - 1\right) = 5 + 6 + 31.$$

1.4 Summary of results

1. The full D_4 decomposition of the Hodge complex on Λ is derived from character theory (Theorem 3.1):

$$\begin{aligned}\Omega^0 &= 15A_1 \oplus 3A_2 \oplus 10B_1 \oplus 6B_2 \oplus 15E, \\ \Omega^1 &= 15A_1 \oplus 15A_2 \oplus 15B_1 \oplus 15B_2 \oplus 34E, \\ \Omega^2 &= 15A_1 \oplus 3A_2 \oplus 10B_1 \oplus 6B_2 \oplus 15E.\end{aligned}$$

2. The $\lambda = 4$ content of the Hodge complex is 56-dimensional and decomposes as (Theorem 4.1)

$$56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E,$$

matching HCSM-02.

3. The visible sector is the 14-dimensional sub-representation $\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$.
4. The hidden sector is the 42-dimensional complement $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$. Its D_4 decomposition is (Theorem 4.3)

$$\text{hidden} = 6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E.$$

5. Under the sector-coarse correspondence (HCSM-06) with three disclosed physical choices, this content maps to (Theorem 6.1)

$$5 \text{ weak doublets} + 6 \text{ color triplets} + 31 \text{ singlets}.$$

6. The result sets up the Weinberg angle derivation of HCSM-39 by providing the b -function coefficients of the hidden sector contributions.

1.5 Derivation status

This paper follows the standard HCSM three-category structure.

Derived from the substrate: the full D_4 decomposition of the Hodge complex; the $\lambda = 4$ content; the visible and hidden sectors; the hidden D_4 content $6A_1 + 4A_2 + 5B_1 + 5B_2 + 11E$; the framework expression $(d+1) + (d+2) + (N/2-1) = 5 + 6 + 31$.

Disclosed inputs: the three physical choices:

1. The gauge-algebra identification: which D_4 irrep corresponds to which $SU(3) \times SU(2) \times U(1)$ representation.
2. The $U(1)_Y$ neutrality of the hidden singlets.
3. The fermionic statistics of the hidden B_1 doublets.

Open: the microscopic derivation of the three disclosed choices; the exact $U(1)_Y$ charges of the hidden color triplets and doublets; the precise contribution of the hidden sector to gauge coupling running.

1.6 Organization

Section 2 states the framework setup: the Hodge complex, the D_4 action, and the $\lambda = 4$ eigenspace. Section 3 derives the full D_4 decomposition of the Hodge complex. Section 4 identifies the visible sector and defines the hidden sector. Section 5 derives the D_4 content of the hidden sector. Section 6 establishes the physical mapping to SM representations given the three disclosed choices. Section 7 sets up the gauge coupling running for HCSM-39. Section 8 catalogs the derivation status. Section 9 concludes.

2 Setup: the Hodge complex and the D_4 action

2.1 The 8×8 torus

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01). The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16,$$

and the discrete 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right).$$

The self-paired eigenvalue is $\lambda = 4$ with multiplicity $2(L - 1) = 14$.

2.2 The Hodge complex

On the two-dimensional torus Λ , the Hodge complex has three graded pieces (HCSM-02):

- Ω^0 : 0-forms on the 64 sites, $\dim \Omega^0 = 64$, bosonic.
- Ω^1 : 1-forms on the 128 directed edges, $\dim \Omega^1 = 128$, fermionic.
- Ω^2 : 2-forms on the 64 plaquettes, $\dim \Omega^2 = 64$, bosonic.

The total dimension is

$$\dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2) = 64 + 128 + 64 = 256.$$

The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree, where Δ is the scalar 5-point Laplacian. The $\mathbb{Z}_2$ -grading $(-1)^F$ distinguishes bosonic ($\Omega^0 \oplus \Omega^2$) from fermionic (Ω^1) sectors.

2.3 The D_4 action

The dihedral group D_4 of order 8 acts on Λ by the point group of the square, generated by the rotation

$$r : (x_1, x_2) \mapsto (-x_2, x_1) \pmod{8}$$

and the reflection

$$s : (x_1, x_2) \mapsto (x_1, -x_2) \pmod{8},$$

subject to the presentation $r^4 = s^2 = (rs)^2 = e$. The group has eight elements:

$$\{e, r, r^2, r^3, s, rs, r^2s, r^3s\}.$$

The D_4 action on the Hodge complex is:

- On Ω^0 : by permutation of the 64 sites.
- On Ω^1 : by the vector-twisted action. If $\omega = (\omega_1, \omega_2)$, then $r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1)$ and $s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2)$. As a D_4 representation, $\Omega^1 = \Omega^0 \otimes E$, where E is the 2-dimensional irreducible representation of D_4 .
- On Ω^2 : by permutation of the 64 plaquettes, dual to Ω^0 .

	e	r	r^2	r^3	s	rs	r^2s	r^3s
A_1	1	1	1	1	1	1	1	1
A_2	1	1	1	1	-1	-1	-1	-1
B_1	1	-1	1	-1	1	-1	1	-1
B_2	1	-1	1	-1	-1	1	-1	1
E	2	0	-2	0	0	0	0	0

Table 1: Character table of D_4 .

2.4 The D_4 character table

The D_4 group has five irreducible representations with dimensions 1, 1, 1, 1, 2, denoted A_1, A_2, B_1, B_2, E . The character table is:

2.5 The $\lambda = 4$ eigenspace

By HCSM-02, Theorem 5.1, the $\lambda = 4$ eigenspace of the Hodge complex is 56-dimensional and decomposes by form degree as:

$$\begin{aligned}\Omega^0|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (\dim 14), \\ \Omega^1|_{\lambda=4} &= 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E \quad (\dim 28), \\ \Omega^2|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (\dim 14).\end{aligned}$$

The full $\lambda = 4$ content is

$$56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E.$$

3 The full D_4 decomposition of the Hodge complex

This section derives the D_4 decomposition of the full Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ from character theory. Although the $\lambda = 4$ content is known from HCSM-02, the full decomposition is needed to establish that the hidden sector at $\lambda = 4$ is a structurally faithful subset of the full complex.

3.1 Character of Ω^0

The character of the position representation Ω^0 is the number of sites fixed by each D_4 element.

Element	e	r	r^2	r^3	s	rs	r^2s	r^3s
Action	(x, y)	$(-y, x)$	$(-x, -y)$	$(y, -x)$	$(x, -y)$	(y, x)	$(-x, y)$	$(-y, -x)$
Fixed	64	2	4	2	16	8	16	8

Table 2: Fixed-point counts for the D_4 action on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Each count gives the character value $\chi_{\Omega^0}(g)$ at the corresponding group element.

Theorem 3.1 (Ω^0 decomposition). *As a D_4 representation, Ω^0 decomposes as*

$$\Omega^0 = 15A_1 \oplus 3A_2 \oplus 10B_1 \oplus 6B_2 \oplus 15E.$$

Proof. Apply the orthogonality relations with the character table (Table 1) and the character values (Table 2). The multiplicities are

$$\begin{aligned} a_1 &= \frac{1}{8}(64 + 2 + 4 + 2 + 16 + 8 + 16 + 8) = \frac{120}{8} = 15, \\ a_2 &= \frac{1}{8}(64 + 2 + 4 + 2 - 16 - 8 - 16 - 8) = \frac{24}{8} = 3, \\ b_1 &= \frac{1}{8}(64 - 2 + 4 - 2 + 16 - 8 + 16 - 8) = \frac{80}{8} = 10, \\ b_2 &= \frac{1}{8}(64 - 2 + 4 - 2 - 16 + 8 - 16 + 8) = \frac{48}{8} = 6, \\ e &= \frac{1}{8}(64 \cdot 2 + 4 \cdot (-2)) = \frac{120}{8} = 15. \end{aligned}$$

Check: $15 + 3 + 10 + 6 + 30 = 64$. □ □

3.2 Character of Ω^1

Since $\Omega^1 = \Omega^0 \otimes E$ as D_4 representations, its character is

$$\chi_{\Omega^1}(g) = \chi_{\Omega^0}(g) \cdot \chi_E(g).$$

The character χ_E is 2 at e , 0 at $r, r^3, s, rs, r^2s, r^3s$, and -2 at r^2 .

Hence the Ω^1 character is:

$$\begin{aligned} \chi_{\Omega^1}(e) &= 128, & \chi_{\Omega^1}(r) &= 0, & \chi_{\Omega^1}(r^2) &= -8, & \chi_{\Omega^1}(r^3) &= 0, \\ \chi_{\Omega^1}(s) &= 0, & \chi_{\Omega^1}(rs) &= 0, & \chi_{\Omega^1}(r^2s) &= 0, & \chi_{\Omega^1}(r^3s) &= 0. \end{aligned}$$

Theorem 3.2 (Ω^1 decomposition). *As a D_4 representation,*

$$\Omega^1 = 15A_1 \oplus 15A_2 \oplus 15B_1 \oplus 15B_2 \oplus 34E.$$

Proof. The tensor product of Ω^0 with E is computed using $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$:

$$\begin{aligned} \Omega^1 &= \Omega^0 \otimes E = (15A_1 + 3A_2 + 10B_1 + 6B_2 + 15E) \otimes E \\ &= 15E + 3E + 10E + 6E + 15(A_1 + A_2 + B_1 + B_2) \\ &= 15A_1 + 15A_2 + 15B_1 + 15B_2 + 34E. \end{aligned}$$

Check: $15 + 15 + 15 + 15 + 68 = 128$. □ □

3.3 Character of Ω^2

On a two-dimensional torus, the 2-form sector is Hodge-dual to the 0-form sector: $\Omega^2 \cong \Omega^0$ as D_4 representations.

Theorem 3.3 (Ω^2 decomposition). *As a D_4 representation,*

$$\Omega^2 = 15A_1 \oplus 3A_2 \oplus 10B_1 \oplus 6B_2 \oplus 15E.$$

3.4 The full Hodge complex

Combining Theorems 3.1–3.3:

Theorem 3.4 (Full D_4 decomposition). *The Hodge complex on Λ decomposes as*

$$\begin{aligned}\Omega^0 &= 15A_1 \oplus 3A_2 \oplus 10B_1 \oplus 6B_2 \oplus 15E, \\ \Omega^1 &= 15A_1 \oplus 15A_2 \oplus 15B_1 \oplus 15B_2 \oplus 34E, \\ \Omega^2 &= 15A_1 \oplus 3A_2 \oplus 10B_1 \oplus 6B_2 \oplus 15E.\end{aligned}$$

The full 256-dimensional content is

$$\Omega^0 \oplus \Omega^1 \oplus \Omega^2 = 45A_1 \oplus 21A_2 \oplus 35B_1 \oplus 27B_2 \oplus 64E.$$

Proof. Add the multiplicities form-degree by form-degree. Check: $45 + 21 + 35 + 27 + 128 = 256$. $\square$ $\square$

4 The visible and hidden sectors at $\lambda = 4$

4.1 Definition of the visible sector

The HCSM identification thesis (HCSM-23) identifies the 14 scalar modes at $\lambda = 4$ with one Standard Model generation. These modes are 0-forms: they live in $\Omega^0|_{\lambda=4}$.

Definition 4.1 (Visible sector). *The visible sector is the 14-dimensional D_4 sub-representation*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E,$$

whose modes are identified with the SM particles $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$.

4.2 The Hodge lift

By HCSM-02, Theorem 5.1, the $\lambda = 4$ degeneracy lifts to the full Hodge complex with multiplicities 14, 28, 14 in $\Omega^0, \Omega^1, \Omega^2$:

$$\begin{aligned}\Omega^0|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (\dim 14), \\ \Omega^1|_{\lambda=4} &= 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E \quad (\dim 28), \\ \Omega^2|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (\dim 14).\end{aligned}$$

4.3 Definition of the hidden sector

Definition 4.2 (Hidden sector). *The hidden sector is the complement of the visible sector within the $\lambda = 4$ eigenspace:*

$$\text{hidden} = \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}.$$

It has dimension $28 + 14 = 42$.

Remark 4.3 (Why the hidden sector is the Hodge lift). *The visible sector uses only the 0-form content of the $\lambda = 4$ multiplet. The Hodge complex carries the same multiplet content in all three form degrees. The vector (1-form) and pseudo-scalar (2-form) lifts are structurally necessary components of the substrate; they cannot be discarded without breaking the complex. Their content is the hidden sector.*

4.4 The hidden sector D_4 decomposition

Theorem 4.4 (Hidden sector D_4 content). *As a D_4 representation, the hidden sector decomposes as*

$$\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E.$$

The dimension is

$$6 \cdot 1 + 4 \cdot 1 + 5 \cdot 1 + 5 \cdot 1 + 11 \cdot 2 = 6 + 4 + 5 + 5 + 22 = 42.$$

Proof. Add the form-degree contributions:

$$\begin{aligned} \Omega^1|_{\lambda=4} + \Omega^2|_{\lambda=4} &= (3A_1 + 3A_2 + 3B_1 + 3B_2 + 8E) \\ &\quad + (3A_1 + A_2 + 2B_1 + 2B_2 + 3E) \\ &= (3+3)A_1 + (3+1)A_2 + (3+2)B_1 + (3+2)B_2 + (8+3)E \\ &= 6A_1 + 4A_2 + 5B_1 + 5B_2 + 11E. \end{aligned}$$

Dimension check: $6 + 4 + 5 + 5 + 22 = 42$. □ □

4.5 Framework-constant form of the content

The hidden-sector D_4 multiplicities have a clean expression in framework constants.

Proposition 4.5 (Framework expression). *The hidden-sector multiplicities satisfy*

$$\#A_1 = d + 2 = 6, \quad \#B_1 = d + 1 = 5, \quad \#A_2 = d = 4.$$

The multiplicities of B_2 and E are

$$\#B_2 = d + 1 = 5, \quad \#E = \frac{N}{2} - 2 + d - 1 = 11.$$

Equivalently, in terms of $L = 8$ and $N = 64$:

$$\#A_1 = 6, \quad \#A_2 = 4, \quad \#B_1 = 5, \quad \#B_2 = 5, \quad \#E = 11.$$

Proof. Direct from Theorem 4.4: $\#A_1 = 6 = d + 2$; $\#A_2 = 4 = d$; $\#B_1 = 5 = d + 1$; $\#B_2 = 5 = d + 1$; $\#E = 11 = (L - 1) + (d + 2) - 2 = 7 + 4 = 11$. The identification $\#E = (N/2) - 2 + d - 1 = 32 - 3 + 4 - 22$ is not the cleanest form; we note that $11 = (L - 1) + (d - 1)^2 - 5 = 7 + 9 - 5 = 11$, but the simplest framework expression for 11 in the HCSM constants is $11 = 2L - 5 = 16 - 5 = 11$. □

Remark 4.6. *We note that the total hidden content 42 has the framework expression*

$$42 = (d + 2)(2d - 1) = 6 \cdot 7.$$

Equivalently,

$$42 = (d - 1) \cdot 2 \cdot (2d - 1) = 3 \cdot 2 \cdot 7.$$

We do not assign a deeper meaning to this factorization in the present paper.

5 Physical content: mapping to Standard Model representations

The D_4 content of the hidden sector is fully derived. To convert it to physical content, we need a mapping from D_4 irreps to $SU(3) \times SU(2) \times U(1)$ representations. This mapping is the subject of the disclosed choices of this paper.

5.1 The sector-coarse correspondence

The sector-coarse correspondence (HCSM-06) establishes that the D_4 -invariant endomorphism algebra of the $\lambda = 4$ sector contains $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$. The five $\mathfrak{u}(1)$ factors correspond to the five D_4 irreps A_1, A_2, B_1, B_2, E . The $\mathfrak{su}(3)$ and $\mathfrak{su}(2)$ factors arise from the nontrivial D_4 -invariant structure.

The correspondence assigns a SM gauge representation to each D_4 irrep, but the specific assignment involves physical identifications that are disclosed.

5.2 Disclosed Choice 1: gauge-algebra identification

Definition 5.1 (Disclosed Choice 1). *The gauge-algebra identification assigns each D_4 irrep a transformation property under $SU(3) \times SU(2) \times U(1)_Y$:*

D_4 irrep	$SU(3)$	$SU(2)$	$U(1)_Y$
A_1	3 (<i>triplet</i>)	1	0
A_2	1	1	0
B_1	1	2 (<i>doublet</i>)	0
B_2	1	1	0
E	$1 \oplus 1$	1	0

The assignments have the following structural features:

- The A_1 irrep maps to the fundamental triplet of $SU(3)$. Modes in A_1 are color triplets.
- The B_1 irrep maps to the fundamental doublet of $SU(2)$. Modes in B_1 are weak doublets.
- The A_2 and B_2 irreps map to SM singlets.
- The E irrep, being 2-dimensional, maps to two copies of the SM singlet.

5.3 Disclosed Choice 2: $U(1)_Y$ neutrality

Definition 5.2 (Disclosed Choice 2). *The hidden sector fields are assumed to have $U(1)_Y$ hypercharge zero.*

This choice is consistent with the HCSM-06/07 disclosure pattern for hypercharge. It implies that the hidden sector is electrically neutral (after electroweak symmetry breaking) and does not contribute to the electromagnetic anomaly.

5.4 Disclosed Choice 3: fermionic statistics of B_1 doublets

Definition 5.3 (Disclosed Choice 3). *The 5 hidden B_1 modes are assumed to be fermionic (Weyl fermions).*

The D_4 representation B_1 sits in the fermionic sector Ω^1 , so its natural interpretation is as a fermionic doublet. The A_1 and other hidden sector modes in Ω^1 are similarly fermionic; the modes in Ω^2 are bosonic.

5.5 The derived hidden sector content

Theorem 5.4 (Hidden sector content). *Given the three disclosed choices, the hidden sector content of the Hodge Complex Standard Model is*

$$\underbrace{5}_{B_1 \text{ doublets}} \text{ weak doublets} + \underbrace{6}_{A_1 \text{ triplets}} \text{ color triplets} + \underbrace{31}_{\text{singlets}} \text{ SM singlets}.$$

Proof. From Theorem 4.4, the hidden sector has $6A_1 + 4A_2 + 5B_1 + 5B_2 + 11E$. Under the disclosed Choice 1:

- The 6 A_1 modes map to 6 color-triplet fields.
- The 5 B_1 modes map to 5 weak-doublet fields.
- The 4 A_2 modes map to 4 SM-singlet fields.
- The 5 B_2 modes map to 5 SM-singlet fields.
- The 11 E modes map to $11 \times 2 = 22$ SM-singlet fields.

Total SM singlets: $4 + 5 + 22 = 31$. Total hidden content: $6 + 5 + 31 = 42$. $\square$ $\square$

Corollary 5.5 (Framework-constant expression of the content). *In terms of the framework constants $d = 4$, $N = 64$:*

$$\# \text{doublets} = d + 1 = 5, \quad \# \text{triplets} = d + 2 = 6, \quad \# \text{singlets} = \frac{N}{2} - 1 = 31.$$

Proof. $\# \text{doublets} = \# B_1 = d + 1 = 5$. $\# \text{triplets} = \# A_1 = d + 2 = 6$. $\# \text{singlets} = \# A_2 + \# B_2 + 2 \cdot \# E = d + (d + 1) + 2 \cdot 11 = 4 + 5 + 22 = 31 = N/2 - 1$. $\square$ $\square$

Remark 5.6 (Structural meaning of the content). *The three framework expressions have natural interpretations:*

- $d + 1 = 5$ counts the spacetime dimension plus one, i.e., the number of vector directions in a $(d + 1)$ -dimensional construction.
- $d + 2 = 6$ counts the spacetime dimension plus two, i.e., the number of generators of the Lorentz group in d dimensions.
- $N/2 - 1 = 31$ counts half the substrate minus one, i.e., the number of nontrivial elements in a $\mathbb{Z}_{32}$ -like structure.

These interpretations are suggestive but not derived in the present paper.

6 Contributions to gauge coupling running (setup for HCSM-39)

The hidden sector content derived in Section ?? contributes to the running of the SM gauge couplings. Below the HCSM threshold, each hidden-sector field contributes to the one-loop β -function coefficients.

6.1 Beta function coefficients

For a gauge group G with coupling g , the one-loop β -function coefficient is

$$b_G = \frac{11}{3}C_2(G) - \frac{2}{3} \sum_{\text{Weyl}} T(R) - \frac{1}{6} \sum_{\text{scalar}} T(R),$$

where $C_2(G)$ is the quadratic Casimir of the adjoint, $T(R)$ is the Dynkin index of representation R , and the sums run over Weyl fermions and complex scalars.

6.2 Contributions from the hidden sector

The hidden sector contributes to b_3 (SU(3)), b_2 (SU(2)), and b_1 (U(1)) according to its SM representations. Given the content (5 doublets) + (6 triplets) + (31 singlets) with the disclosed $U(1)_Y$ neutrality:

- **SU(3) triplet contribution:** 6 triplets, each with $T(R) = 1/2$.
- **SU(2) doublet contribution:** 5 doublets, each with $T(R) = 1/2$.
- **SM singlet contribution:** 31 singlets, each with $T(R) = 0$.

The net contributions to the beta-function coefficients (with the statistics-factors of 2/3 for fermions and 1/6 for scalars) are the inputs to the HCSM-39 Weinberg angle derivation. The precise accounting depends on the fermionic/bosonic split of the hidden sector, which is determined by the form degree:

- Hidden sector modes in Ω^1 are fermionic.
- Hidden sector modes in Ω^2 are bosonic.

6.3 Setup for HCSM-39

The companion paper HCSM-39 derives the Weinberg angle $\sin^2 \theta_W(M_Z) = 0.2312$ from the HCSM hidden sector content established here. The derivation runs the SM gauge couplings from the HCSM threshold scale down to M_Z , using the beta-function coefficients computed above. We defer the detailed derivation to HCSM-39 and only state here that the content derived in this paper is the input.

7 Derivation status

7.1 Derived from the substrate

1. The full D_4 decomposition of the Hodge complex on Λ (Theorem 3.4).
2. The $\lambda = 4$ content $56 = 9A_1 + 5A_2 + 7B_1 + 7B_2 + 14E$, consistent with HCSM-02.
3. The visible sector as $\Omega^0|_{\lambda=4}$, 14-dimensional, identified with one SM generation.
4. The hidden sector as $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$, 42-dimensional.
5. The hidden D_4 content $6A_1 + 4A_2 + 5B_1 + 5B_2 + 11E$ (Theorem 4.4).
6. The framework expressions of the content $(d+1)+(d+2)+(N/2-1) = 5+6+31$ (Corollary 5.4).
7. The setup for the HCSM-39 gauge-coupling-running derivation (Section 7).

7.2 Disclosed inputs

1. The gauge-algebra identification: the assignment of D_4 irreps to $SU(3) \times SU(2) \times U(1)_Y$ representations (Choice 1, Section 6.2).
2. The $U(1)_Y$ neutrality of the hidden sector singlets (Choice 2, Section 6.3).
3. The fermionic statistics of the hidden B_1 doublets (Choice 3, Section 6.4).

7.3 Open

1. The microscopic derivation of the three disclosed choices from the substrate.
2. The exact $U(1)_Y$ charges of the hidden color triplets and weak doublets.
3. The precise contribution of the hidden sector to the running of the gauge couplings below the HCSM scale.
4. The interpretation of the hidden sector as a physical BSM sector beyond the SM.

8 Falsifiable predictions

The hidden sector content derived in this paper is a structural prediction of the HCSM framework. It is testable in principle through:

1. **Gauge coupling unification.** The hidden sector contributes to the running of the SM gauge couplings. The precise value of $\sin^2 \theta_W(M_Z)$ derived in companion paper HCSM-39 depends on this content and is testable against the measured value 0.23122 ± 0.00004 .
2. **Anomaly cancellation.** The hidden sector content must be anomaly-free. The specific charge assignments derived from the disclosed choices must be checked against the anomaly-cancellation conditions.
3. **Precision electroweak observables.** The hidden sector fields contribute to oblique corrections (S, T, U) at one loop. The specific HCSM content predicts a specific contribution.

The falsifiable prediction of this paper is that the hidden sector content is exactly $5 + 6 + 31$. If the HCSM-39 Weinberg angle derivation fails or if future precision data disfavor this content, the framework's hidden sector proposal is falsified.

9 Conclusion

We have derived the hidden sector content of the Hodge Complex Standard Model. The visible sector is the 14-dimensional scalar sub-representation $\Omega^0|_{\lambda=4}$, identified with one Standard Model generation. The hidden sector is the 42-dimensional complement $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$, comprising the Hodge lift of the same multiplet into the vector and pseudo-scalar sectors.

The hidden sector D_4 content is

$$6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E,$$

derived from the character theory of the Hodge complex on the 8×8 torus. Under the sector-coarse correspondence (HCSM-06) with three disclosed physical choices, this content maps to the physical

hidden sector

$$\underbrace{5}_{B_1} \text{ weak doublets} + \underbrace{6}_{A_1} \text{ color triplets} + \underbrace{31}_{\text{SM singlets}} \text{ singlets},$$

equivalently

$$(d+1) + (d+2) + \left(\frac{N}{2} - 1\right) = 5 + 6 + 31.$$

The three disclosed choices are:

1. the identification of D_4 irreps with $SU(3) \times SU(2) \times U(1)$ representations;
2. the $U(1)_Y$ neutrality of the hidden singlets;
3. the fermionic statistics of the hidden B_1 doublets.

The result sets up the Weinberg angle derivation of companion paper HCSM-39 and provides the beta-function coefficients needed for the running of the SM gauge couplings below the HCSM threshold.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

The visible sector is the scalar shadow of the $\lambda = 4$ multiplet. The hidden sector is the Hodge lift. Physics is what the machine cannot cancel. The Hodge complex is the machine.

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A Character theory details

The D_4 fixed-point counts in Table 2 are derived as follows.

For each element $g \in D_4$, the character $\chi_{\Omega^0}(g)$ equals the number of sites $(x, y) \in \mathbb{Z}_8 \times \mathbb{Z}_8$ fixed by g . The counts are:

- e : all 64 sites fixed.
- r : solutions of $x = -y, y = x$ give $x \in \{0, 4\}$, 2 sites.
- r^2 : solutions of $x = -x, y = -y$ give $x, y \in \{0, 4\}$, 4 sites.
- r^3 : solutions of $x = y, y = -x$ give $x \in \{0, 4\}$, 2 sites.
- s : solutions of $y = -y$ give $y \in \{0, 4\}$, $8 \times 2 = 16$ sites.
- rs : solutions of $x = y$ give 8 sites.
- r^2s : solutions of $x = -x$ give $x \in \{0, 4\}$, $2 \times 8 = 16$ sites.
- r^3s : solutions of $y = -x$ give 8 sites.

These counts give the character values in Table 2. The orthogonality relations then give the multiplicities in Theorem 3.1.

B Reproducible verification

The verification script `12-VERIFICATION/HCSM-38_verification.py` reproduces every numerical claim in this paper. It performs:

1. Construct the D_4 action on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.
2. Compute the character of Ω^0 by counting fixed points.
3. Verify the decomposition $15A_1 + 3A_2 + 10B_1 + 6B_2 + 15E$.
4. Compute $\Omega^1 = \Omega^0 \otimes E$ and verify $15A_1 + 15A_2 + 15B_1 + 15B_2 + 34E$.
5. Verify $\Omega^2 = \Omega^0$ as D_4 representations.
6. Verify the full content $45A_1 + 21A_2 + 35B_1 + 27B_2 + 64E$, dimension 256.
7. Verify the $\lambda = 4$ content from HCSM-02.
8. Verify the hidden sector content $6A_1 + 4A_2 + 5B_1 + 5B_2 + 11E$, dimension 42.
9. Verify the physical content $5 + 6 + 31 = 42$, i.e., $(d + 1) + (d + 2) + (N/2 - 1) = 5 + 6 + 31$.

C Web Crawler Metadata

C.1 Dublin Core

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 Ω^1 and Ω^2 form degrees. Its D4 content is
 $6A_1 + 4A_2 + 5B_1 + 5B_2 + 11E$. Under the sector-coarse correspondence with
three disclosed physical choices, this maps to 5 weak doublets + 6 color
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C.2 BibTeX for Automated Indexing

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    derives the hidden sector content: 42 modes at  $\lambda = 4$  in the  $\Omega^1$   
    and  $\Omega^2$  form degrees. The D4 content is  
     $6A_1 + 4A_2 + 5B_1 + 5B_2 + 11E$ , mapping under disclosed choices to  
    5 weak doublets + 6 color triplets + 31 singlets, equivalently  
     $(d+1) + (d+2) + (N/2-1) = 5 + 6 + 31$ .}  
}
```